

Itaru Ota

Full-stack Software Engineer - Cloud Infrastructure and AI-enabled Systems

Based in Tokyo, Japan

- [GitHub](#)
- [Portfolio](#)
- [LinkedIn](#)

Profile

Full-stack software engineer experienced in product development across frontend, backend, data models, APIs, and cloud infrastructure. Builds and operates systems with TypeScript, Python, Terraform, AWS, and Google Cloud, with an emphasis on maintainable architecture, reproducible delivery, and measurable performance. Background in natural language processing and applied machine learning.

Technical Skills

- **Languages:** Python, TypeScript, Java, Rust
- **Application development:** Django, gRPC, NestJS, Spring Boot, React, Next.js
- **Cloud and infrastructure:** AWS, Azure, Google Cloud, Terraform, Docker, GitHub Actions
- **Data and AI:** PostgreSQL, MySQL, Redis, Prisma, Azure OpenAI Service, PyTorch, OCR, spaCy
- **Architecture and quality:** API and data-model design, DDD, CI/CD, pytest, Jest

Experience

3-shake Inc. - Full-stack Engineer

Jan 2025 - Present

- Deliver features for a B2B SaaS product across a Next.js frontend, Django and gRPC backends, data models, and cloud infrastructure
- Design, implement, and operate containerized services on Amazon ECS, including service networking and infrastructure provisioned with Terraform
- Automate deployments and end-to-end testing through CI/CD, and establish observability and incident-response practices using logs, traces, and metrics
- Build performance benchmarks to support architecture decisions, including evaluation of a Python-to-Rust service migration
- Create reusable technical training materials and serve as an instructor and mentor for engineers

Daiwa Institute of Research Ltd. - Application Engineer

Apr 2023 - Dec 2024

- Developed and modernized enterprise systems from requirements and design through release and operations
- Designed and implemented business APIs and domain models using Java and Spring, and developed web applications on AWS
- Built and evaluated a retrieval-augmented generation system using Python and Azure OpenAI Service
- Applied natural language processing and machine learning to economic-data analysis and forecasting

Experience (continued)

SOAT Inc. - Part-time Software Engineer

Oct 2021 - Mar 2023

- Developed OCR applications that converted PDF documents into structured, editable data using Python, OpenCV, and Tesseract
- Improved object-detection models for manufacturing inspection through data preparation, augmentation, and model evaluation
- Built reproducible development environments and evaluation tools with Docker, SQLite, MySQL, and PyTorch

Open Source

[limco - Japanese Text Analysis Library](#)

- Contributed to an open-source authorship-analysis library developed at the NAIST Social Computing Laboratory
- Implemented Japanese natural language processing pipelines and linguistic metrics with spaCy and GiNZA

Community and Recognition

[2026 Japan AWS Jr. Champion](#)

Selected as one of 114 early-career engineers in Japan for technical capability, organizational and community contributions, and public knowledge sharing. Published technical articles on Amazon ECS networking and Terraform-based Infrastructure as Code.

Presentations (2025-2026)

- [NW-JAWS #22: “Reaching Services on Service Connect from Outside an ECS Service”](#), Aug 2026
- [JAWS-UG Ibaraki #15: “Continuing to Engage with Software Engineering”](#), Jul 2026
- [Meguro LT #37: “Reading the Why Behind Commits”](#), Jun 2026
- [3-shake SRE Tech Talk LT](#), Sep 2025
- [JAWS-UG Container Branch LT](#), Aug 2025
- [Qiita Event: “Bash - Let’s Talk About Recent AI Use Cases”](#), May 2025

Certifications

- [AWS Certifications](#), 2023-2026
- Google Cloud Associate Cloud Engineer and Generative AI Leader, 2025
- Applied Information Technology Engineer Examination, 2023
- TOEIC Listening and Reading: 865, 2023

Education and Research

- **M.Eng. in Information Science**, Nara Institute of Science and Technology, Mar 2023
- **B.A. in Economics**, Keio University, Mar 2021

Graduate research focused on improving the experience of listening to broadcast programs through social-media analysis. Presented the work at DEIM 2022 and DEIM 2023 and received the [DEIM 2022 Student Presentation Award](#) and the [imedia2021 Data Broad Award](#).